

ASSESSMENT TOOL - 1

Course code	CSA0702	Course Title	Computer Networks
Co Assessed	Co 1	Assessment	Tool - Short Answers
Weightage		Total marks	20 marks

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Section : Batch	IT - 2024	Date of Submission:	17/07/2024
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Total marks	20	Obtained marks	17
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ASSESSMENT TOOL-1

SHORT ANSWERS

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Course name: Computer Networks

Course code: CSA7102

1. Transport Layer + Data link layer

Given

file size = 150 MB

Bandwidth = 50 Mbps

Segment size = 1500 bytes

Header + Trailer = 40 bytes/frame

Step 1 : No. of Segments

Using 1 MB = 1,000,000 bytes 150 MB = 150,000,000 bytes

$$\text{Segments} = \frac{150,000,000}{1500} = 100,000$$

$$\text{Segments} = 100,000$$

Step 2 No. of frames

Each segment becomes one frame

$$\text{frames} = 100,000$$

Step 3 Total Overhead

$$100,000 \times 40 = 4,000,000 \text{ bytes} = 4 \text{ MB}$$

Step 4 Transmission time

$$150 \text{ Mb} = 1200 \text{ Mb}$$

$$\frac{1200}{50} = 24 \text{ sec}$$

Transmission time = 24 Seconds

2)

Sliding Window

Given

Window size = 6

Total frames = 48

Frame size = 1024 bytes

Step 1

$$\frac{48}{6} = 8$$

Transmission windows = 8

Step 2

one frame lost in every window

$$\text{Retransmissions} = 8 \times 1 = 8$$

Theory:

flow control prevents the sender from overwhelming the receiver, reducing congestion & improving efficiency

3)

IP Fragmentation

Packet = 12,000 bytes

MTU = 1500 bytes

Header = 20 bytes

Step 1

Each fragment can carry

$$1500 - 20 = 1480 \text{ bytes}$$

Step 2

$$\frac{12000}{1480} = 8.1$$

Need 9 fragments

Step 3

Header overhead

$$9 \times 20 = 180 \text{ bytes}$$

Theory:

The Network layer fragments large packets and reassembles them at the destination using Identification numbers and fragment offsets.

4) Sliding Window

$$\text{Transmission windows} = 48/6 = 8$$

$$\text{Retransmissions} = 8$$

Theory:

Flow control regulates transmission speed and avoids receiver buffer overflow.

5) Session layer:

Given

$$\text{Video} = 600 \text{ MB}$$

Checkpoint every 60 MB

failure after 390 MB

Step 1

Checkpoints before failure

60, 120, 180, 240, 300, 360

Total Checkpoints = 6

Step 2

Recovery starts from 360 MB

Retransmitted data

$$390 - 360 = 30 \text{ MB}$$

$$\text{Retransmission} = 30 \text{ MB}$$

Theory: Synchronization checkpoints allow communication to resume from the last checkpoint instead of transmitting the entire file.

6. Presentation layer:

Given

Original = 900 MB

Compression = 40%

Encryption overhead = 5%

Step 1

compressed size

$$900 \times 0.60 = 540 \text{ MB}$$

Step 2

Encrypted size

$$540 \times 1.05 = 567 \text{ MB}$$

Step 3

Reduction

$$900 - 567 = 333 \text{ MB}$$

$$\frac{333}{900} \times 100 = 37\%$$

Theory:

The presentation layer compresses data to reduce transmission size and encrypts it for confidentiality.

7) FTP Application

Given

Data = 3 GB

Throughput = 120 Mbps

Request Size = 3 MB

Step 1

$$3 \text{ GB} = 3000 \text{ MB}$$

Requests

$$3000/3 = 1000$$

$$\text{Request} = 1000$$

Step 2

$$3 \text{ GB} = 24000 \text{ Mb}$$

Transmission time

$$\frac{24000}{120} = 200 \text{ sec}$$

Theory:

The application layer provides services like FTP file transfer, authentication and user interaction.

8) End to End Delay

Given

$$\text{Application} = 4 \text{ ms}$$

$$\text{Transport} = 3 \text{ ms}$$

$$\text{Network} = 5 \text{ ms}$$

$$\text{Data link} = 2 \text{ ms}$$

$$\text{Physical} = 1 \text{ ms}$$

$$\text{propagation} = 18 \text{ ms}$$

$$\text{Transmission} = 12 \text{ ms}$$

$$\text{Total} = 4 + 3 + 5 + 2 + 1 + 18 + 12 = 45 \text{ ms}$$

$$\text{Total delay} = 45 \text{ ms}$$

Theory:

Each OSI layer performs specific processing before data is transmitted propagation and transmission

delays contribute to the overall communication time

9) protocol overhead

Useful data = 80 MB

frame size = 1600 bytes

Overhead = 80 bytes

Step 1

frames

$$80,000,000 / 1600 = 50,000$$

Step 2

Total overhead

$$50,000 \times 80 = 4,000,000 \text{ bytes} = 4 \text{ MB}$$

Step 3

Efficiency

Useful payload / frame

$$1600 - 80 = 1520$$

$$\frac{1520}{1600} \times 100 = 95\%$$

Theory:

Higher protocol overhead reduces bandwidth efficiency because more bytes are used for headers and trailers instead of user data.

10) Multiple OSI layers

Given

Original file = 240 MB

Compression = 30%

Segment size = 1400 bytes

frame overhead = 60 bytes

Step 1

$$\text{Compressed size} = 240 \times 0.70 = 168 \text{ MB}$$

Step 2

Compressed bytes

168,000,000

Segments

$$\frac{168,000,000}{1400} = 120,000$$

$$\text{Segment} = 120,000$$

Step 3

Protocol overhead

$$120,000 \times 60 = 7,200,000 \text{ bytes}$$

$$= 7.2 \text{ MB}$$

Step 4

Final transmitted Data

$$168 + 7.2 = 175.2 \text{ MB}$$

Theory:

presentation layer : compresses & encrypts data

Transport layer : Segments data and ensures reliable delivery

Data link layer : Adds frame headers / trailers for error detection

Physical Layer : Transmits the final bits over the communication medium.

Q.11
20/11/20

Understanding of concept	Explanation	Application	Organization	Accuracy	Clarity.
25%	25%	20%	15%	10%	5%
5	5	4	2	0.5	0.5